



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Project Objective**

- Develop a machine learning model capable of predicting whether the Falcon 9 first stage will land successfully using historical SpaceX launch data.

- **Project Workflow**

- Data collection using the SpaceX REST API and Wikipedia web scraping
- Data wrangling and preprocessing
- Exploratory data analysis (EDA)
- SQL analysis
- Interactive visualization with Folium and Plotly Dash
- Machine learning model development and evaluation

- **Key Outcome**

- The best-performing classification models achieved a **test accuracy of 83.33%**, demonstrating that launch site, payload mass, and orbit type are important predictors of landing success.

Introduction

- **Background**

SpaceX significantly reduces launch costs by recovering and reusing Falcon 9 first-stage boosters.

Accurately predicting landing success helps improve:

- Mission planning
- Operational efficiency
- Cost estimation
- Launch reliability

- **Project Goal**

Build a predictive classification model that estimates whether a Falcon 9 booster will successfully land.

Section 1

Methodology

Methodology

SpaceX API + Wikipedia



Data Collection



Data Wrangling



EDA & SQL Analysis



Interactive Analytics
(Folium & Plotly Dash)



Machine Learning



Landing Success Prediction

Data Collection

- **Objective**

Collect historical Falcon 9 launch information from multiple reliable sources.

- **Data Sources**

- **SpaceX REST API**

- Launches
- Rockets
- Payloads
- Launch Pads
- Booster Information

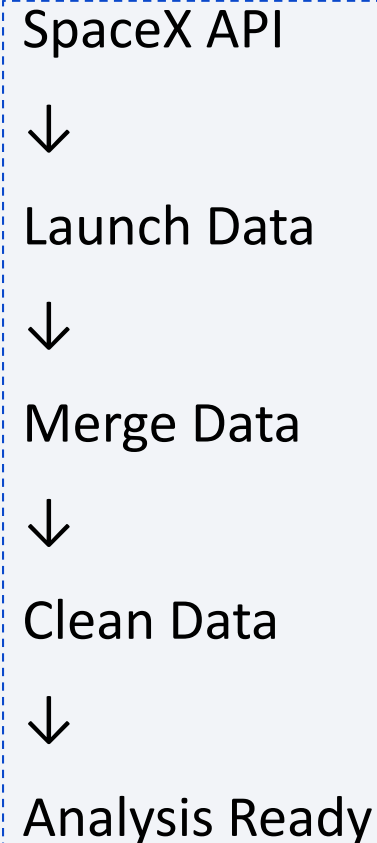
- **Wikipedia**

- Falcon 9 launch history
- Mission outcomes
- Landing outcomes

<https://github.com/Frqmrg/Frqmrg>

Data Collection – SpaceX API

- **Web Scraping from Wikipedia**
- **Objective**
 - Collect historical Falcon 9 launch records unavailable through the API.
- **Methodology**
 - Retrieved webpage using Requests.
 - Parsed HTML tables with BeautifulSoup.
 - Extracted launch date, payload, orbit, launch site, mission outcome, and landing outcome.
 - Converted extracted data into a Pandas DataFrame.
- **Output**
 - Historical launch dataset
 - Structured tabular data
 - Dataset ready for preprocessing



```
graph TD; A[SpaceX API] --> B[Launch Data]; B --> C[Merge Data]; C --> D[Clean Data]; D --> E[Analysis Ready];
```

SpaceX API

↓

Launch Data

↓

Merge Data

↓

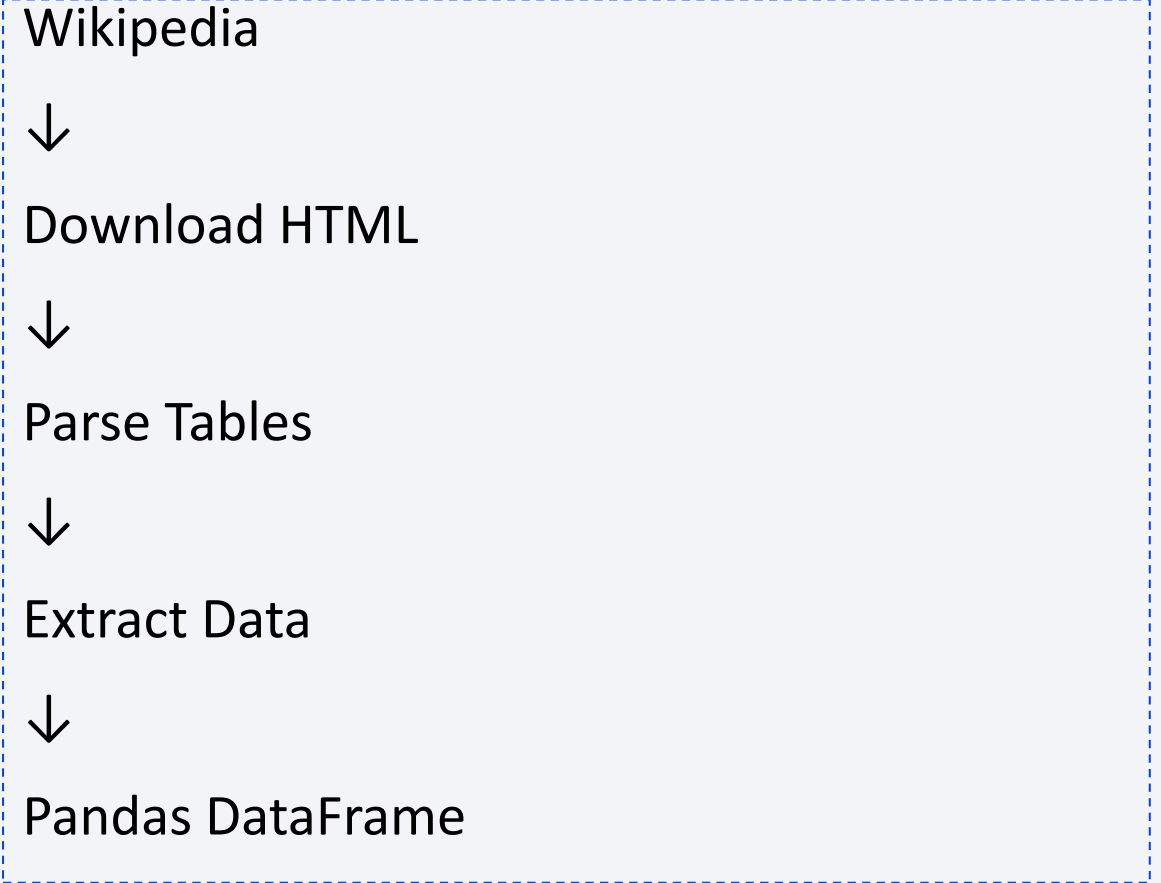
Clean Data

↓

Analysis Ready

Data Collection - Scraping

- **Objective**
- Collect historical Falcon 9 launch records unavailable through the API.
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Data Wrangling

Data Preparation

Tasks Performed

- Removed duplicate records
- Handled missing values
- Converted categorical variables
- Created landing success labels
- Standardized numerical features

Output

- Machine-learning-ready dataset for predictive modeling.

Raw Dataset



Data Cleaning



Feature Engineering



Encoding



ML Dataset

EDA with Data Visualization

Purpose

Understand relationships between launch characteristics and landing success.

Visualizations Created

- Flight Number vs Launch Site
- Payload vs Launch Site
- Success Rate by Orbit
- Flight Number vs Orbit
- Payload vs Orbit
- Yearly Launch Success Trend

Outcome

EDA identified launch site, payload, and orbit type as important predictive features.

EDA with SQL

SQL Queries Performed

- Retrieve unique launch sites
- Calculate payload statistics
- Identify successful landing records
- Determine maximum payload boosters
- Rank landing outcomes
- Analyze mission success by year

Key Finding

- SQL analysis revealed meaningful trends in launch locations, payload capacity, and landing outcomes.

<https://github.com/Frqmrg/Frqmrg>

Build an Interactive Map with Folium

Interactive Geographic Analysis

Features Implemented

- Launch site markers
- Landing outcome color coding
- Distance to highways
- Distance to coastlines
- Distance to railways
- Airport proximity

Insight

- Geographic visualization helps identify operational characteristics surrounding each launch site.

Build a Dashboard with Plotly Dash

Dashboard Components

- Launch Site Filter
- Success Pie Chart
- Payload Range Slider
- Payload vs Landing Outcome Scatter Plot

Purpose

- Enable interactive exploration of launch performance across different launch sites and payload ranges.
- GitHub Repository

Predictive Analysis (Classification)

Machine Learning Workflow

Models Developed

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)

Model Optimization

- Feature scaling
- Train/Test split
- Hyperparameter tuning using GridSearchCV
- 10-fold Cross Validation

Evaluation Metric

- Classification Accuracy on the test dataset.

Dataset



Train/Test Split



Train Models



GridSearchCV



Best Model

Results

Exploratory Data Analysis (EDA)

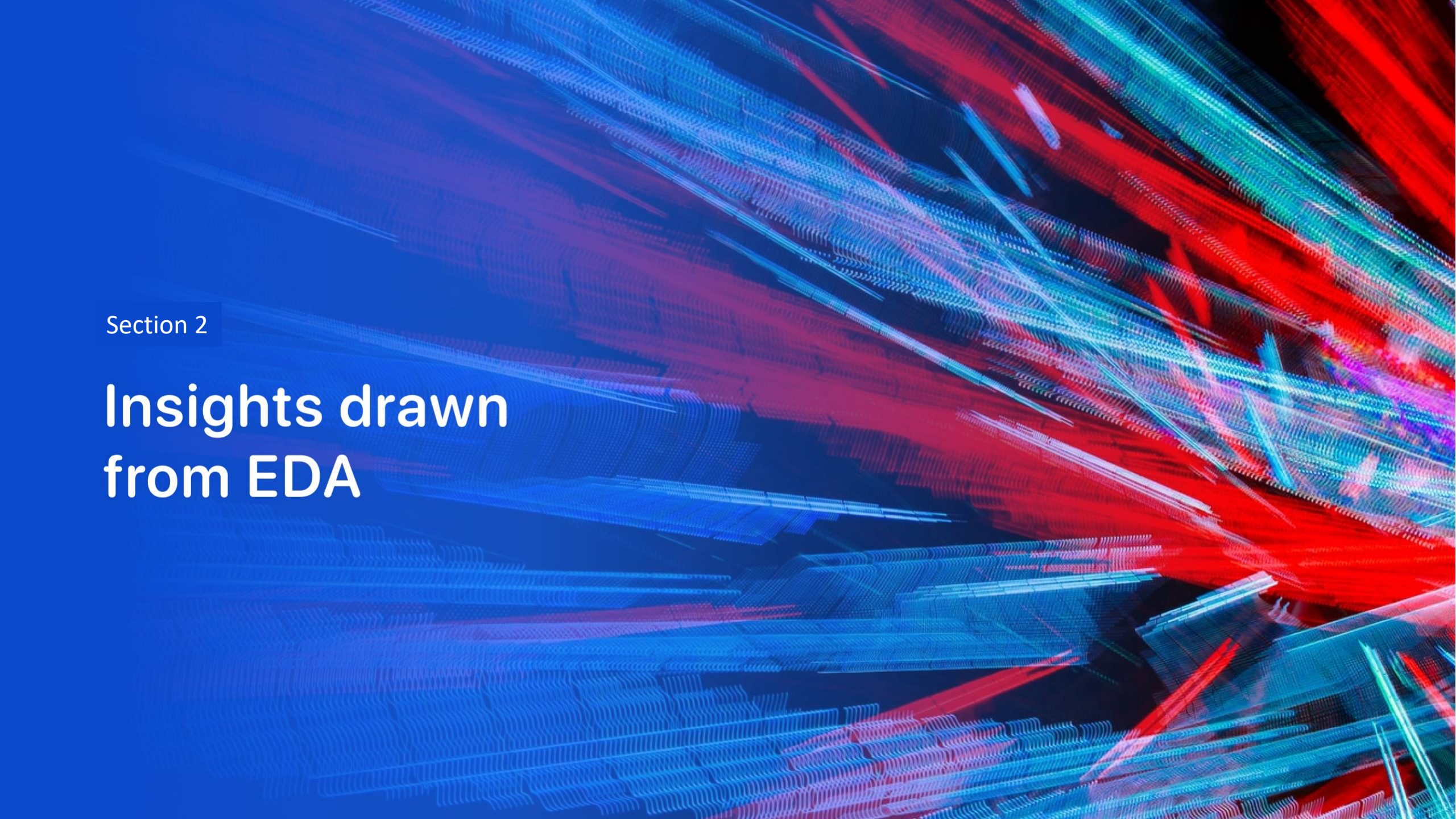
- Visual analysis of launch characteristics and landing outcomes.
- Identification of trends related to launch site, payload mass, and orbit type.

Interactive Analytics

- Geographic visualization using **Folium Maps**.
- Interactive dashboard development using **Plotly Dash**.

Predictive Analysis

- Development and evaluation of machine learning classification models.
- Comparison of Logistic Regression, SVM, Decision Tree, and KNN.
- Best classification accuracy achieved: **83.33%**.

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Objective

- Analyze how launch frequency varies across different launch sites.

Key Observations

- Launch frequency increased steadily over time.
- **KSC LC-39A** achieved a high proportion of successful landings.
- Later missions showed improved landing success, reflecting advancements in booster recovery.



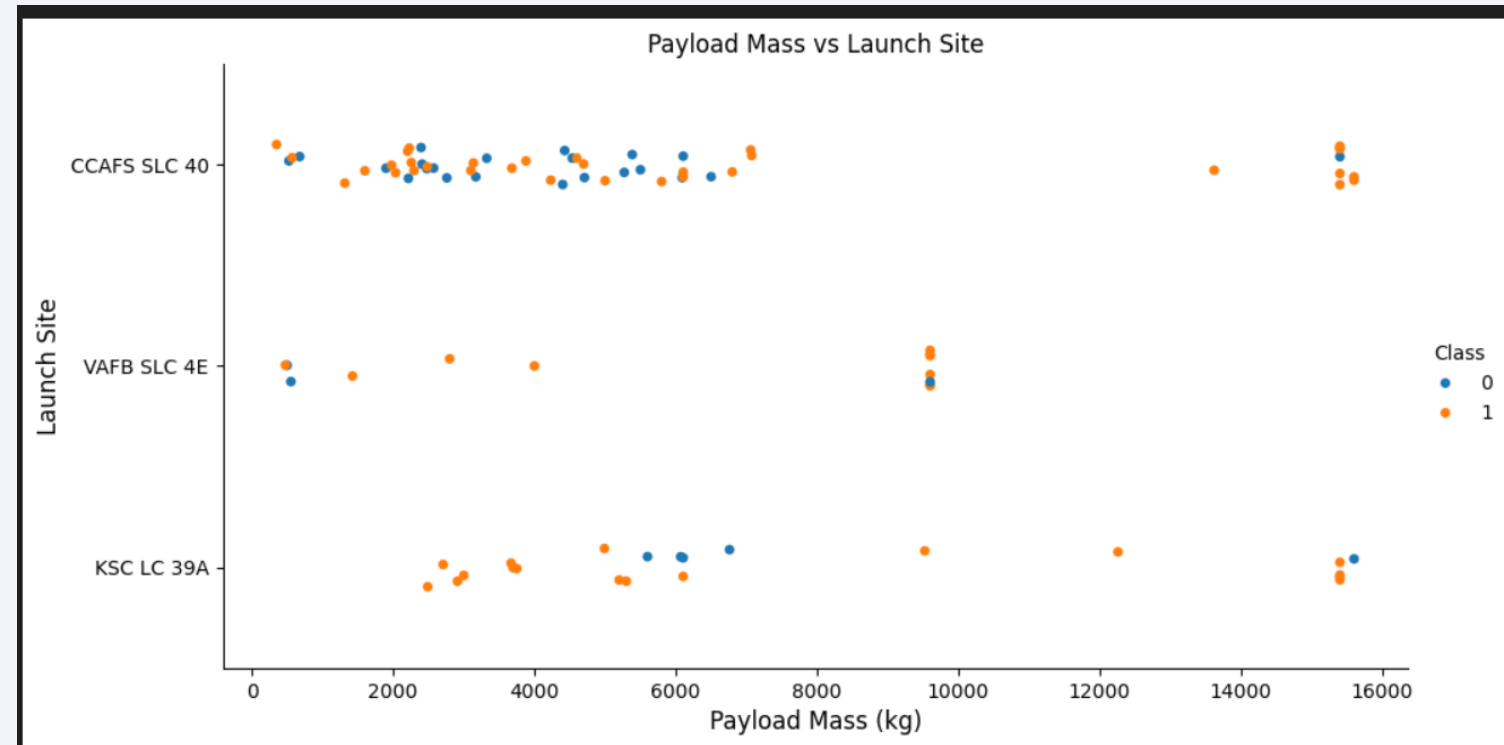
Payload vs. Launch Site

- **Objective**

Examine the relationship between payload mass and launch site.

Key Observations

- Launch sites handled a wide range of payload masses.
- Higher payload missions were successfully launched from multiple sites.
- Launch success was influenced by both payload mass and launch location.



Success Rate vs. Orbit Type

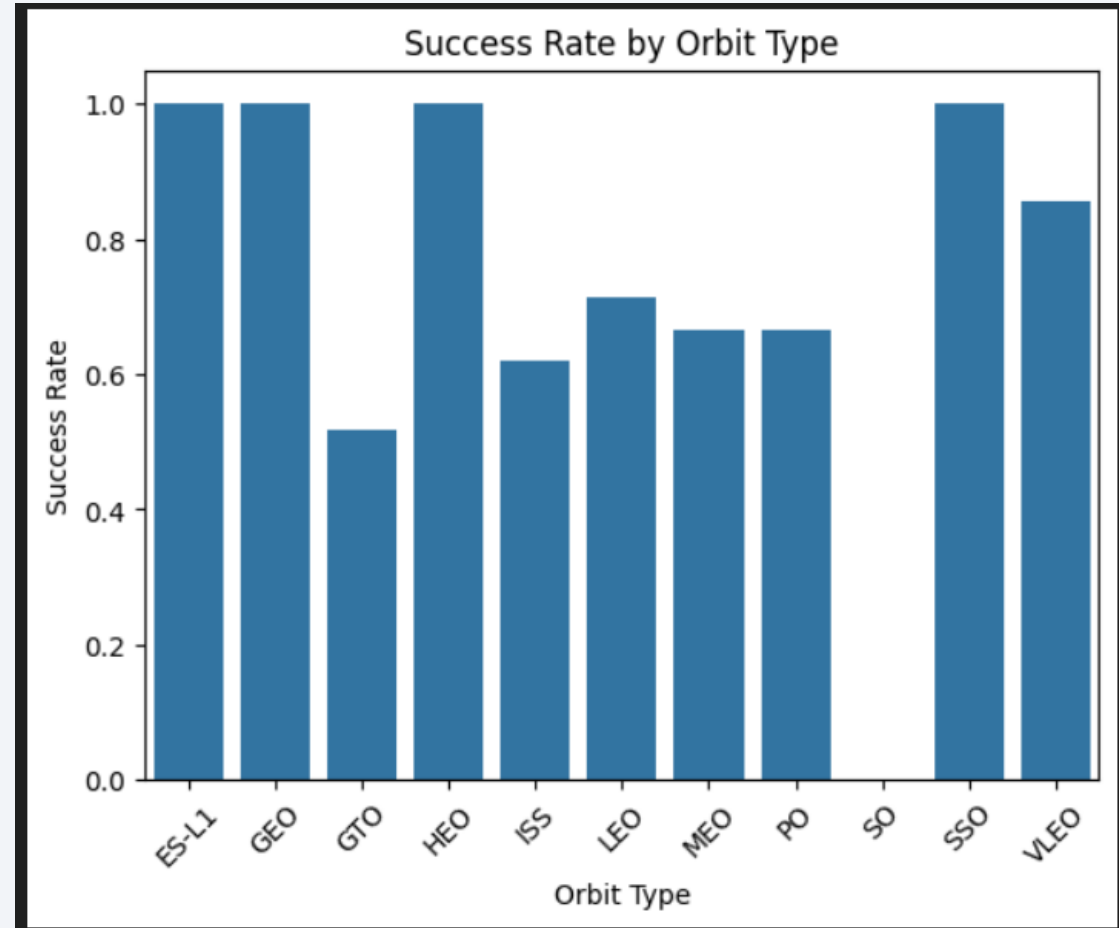
Objective

- Compare landing success rates across different orbit types.

Key Observations

Success rates varied significantly by orbit.

- Orbit type is an important predictor of landing outcome.



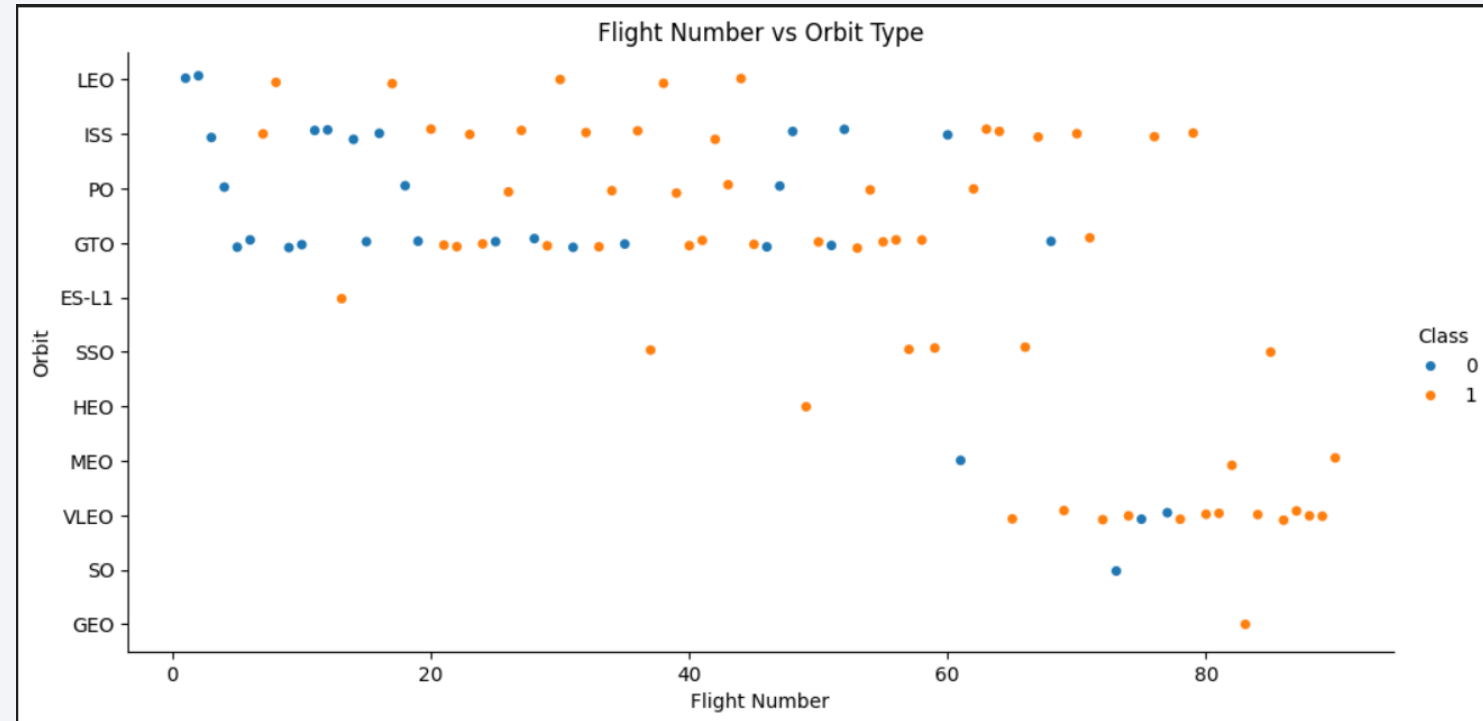
Flight Number vs. Orbit Type

Objective

- Evaluate how mission sequence relates to orbit type.

Key Observations

- Early launches showed more unsuccessful landings.
- Later flight numbers demonstrated improved landing success.
- Recovery performance improved with operational experience.



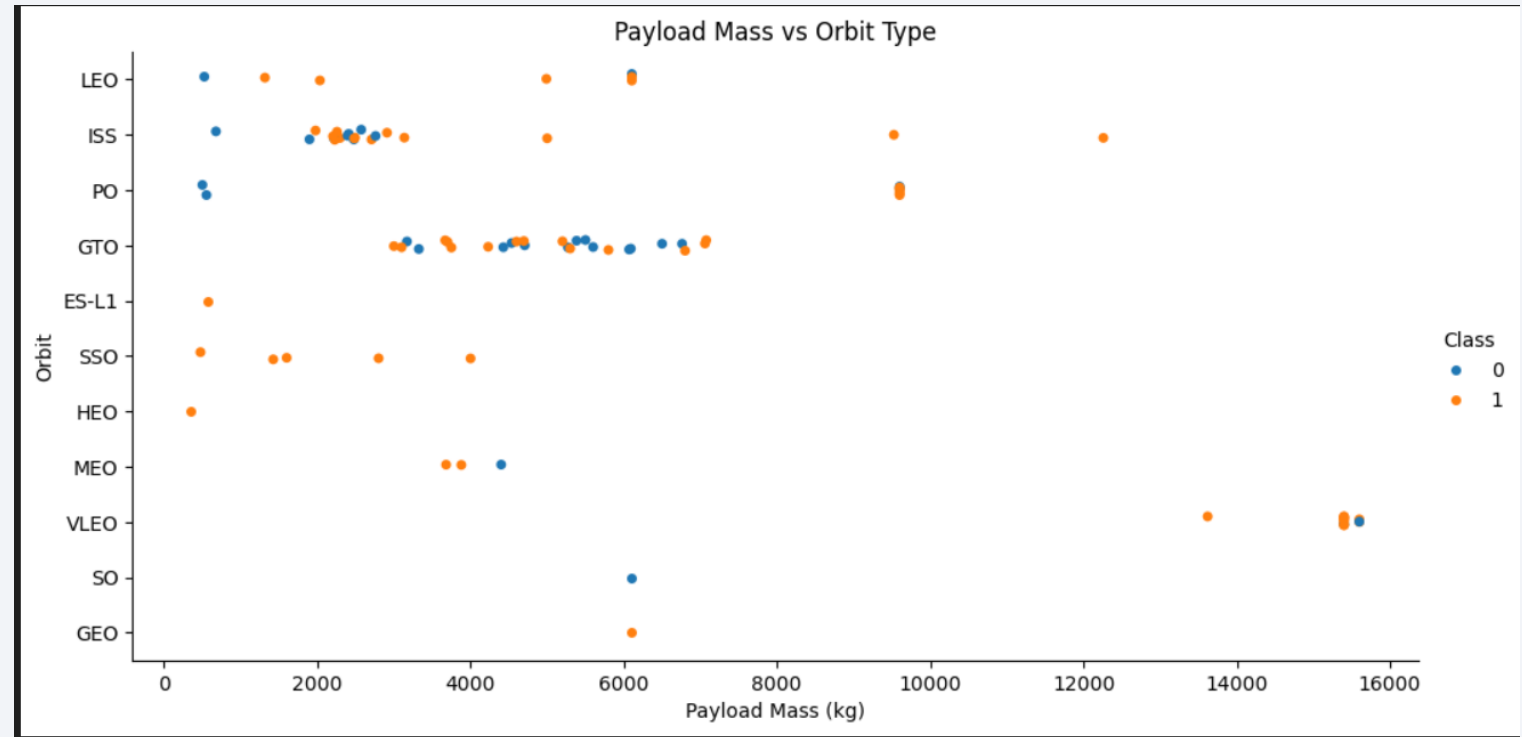
Payload vs. Orbit Type

Objective

- Study the relationship between payload mass and orbit.

Key Observations

- Payload mass differed across orbit categories.
- Certain orbit types consistently carried heavier payloads.
- Payload characteristics contributed to landing success.



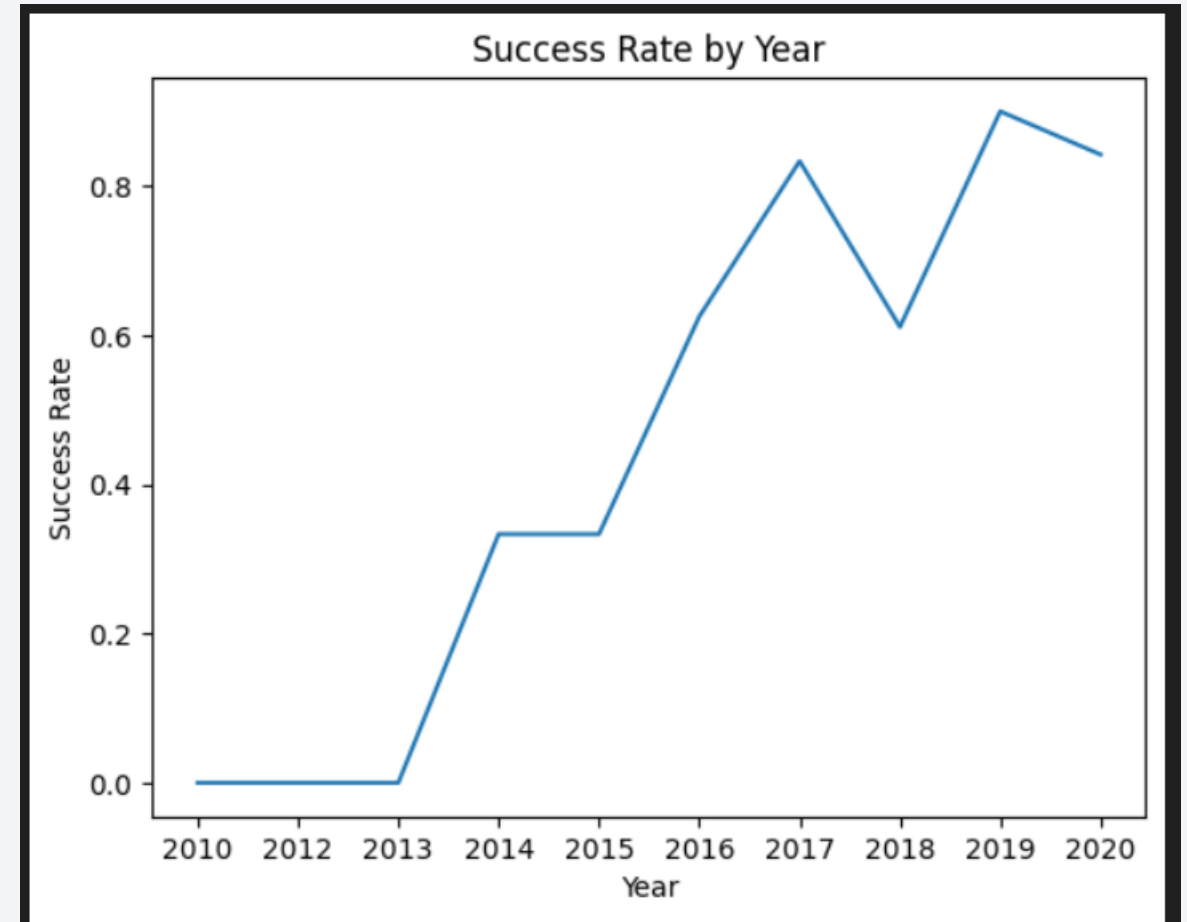
Launch Success Yearly Trend

Objective

- Analyze yearly improvements in Falcon 9 landing success.

Key Observations

- Launch success increased steadily over time.
- Significant improvement occurred after booster reuse technology matured.
- Later years achieved consistently higher success rates.



All Launch Site Names

SQL Query Objective

- Retrieve all unique launch sites.

Result Summary

- The dataset contains four primary launch sites:
- CCAFS LC-40
- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

```
%%sql  
SELECT DISTINCT Launch_Site  
FROM SPACEXTABLE;
```

Launch Site Names Begin with 'CCA'

SQL Query Objective

- Identify launch sites starting with **CCA**.

Observation

- The query confirmed multiple launches from Cape Canaveral facilities, highlighting their importance in Falcon 9 operations.

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Total Payload Mass

SQL Query Objective

- Calculate the cumulative payload mass for NASA missions.

Observation

- NASA missions accounted for a substantial portion of the total payload mass carried by Falcon 9.

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS
Total_Payload_Mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Average Payload Mass by F9 v1.1

SQL Query Objective

- Calculate the average payload mass for **Falcon 9 v1.1**.

Observation

- The result provides insight into the lifting capability of the Falcon 9 v1.1 booster configuration.

```
%%sql
SELECT AVG(PAYLOAD_MASS__KG_) AS
Average_Payload_Mass
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

First Successful Ground Landing Date

SQL Query Objective

- Identify the first successful ground landing.

Observation

- This milestone marked a major advancement in reusable rocket technology.

```
%%sql
SELECT MIN(Date) AS
First_Successful_Ground_Pad_Landing
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

Successful Drone Ship Landing with Payload between 4000 and 6000

SQL Query Objective

- Retrieve successful drone ship landings with payloads between **4000–6000 kg**.

Observation

- The query demonstrates Falcon 9's capability to recover boosters while carrying medium-weight payloads.

```
%%sql
SELECT DISTINCT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success
(drone ship)'
AND PAYLOAD_MASS__KG_ > 4000
AND PAYLOAD_MASS__KG_ < 6000;
```


Total Number of Successful and Failure Mission Outcomes

SQL Query Objective

- Count successful and failed missions.

Observation

- Successful missions significantly outnumbered failures, reflecting continuous improvements in launch operations.

```
%%sql
SELECT Mission_Outcome,
       COUNT(*) AS Total_Missions
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

Boosters Carried Maximum Payload

SQL Query Objective

- Identify the booster that carried the maximum payload.

Observation

- This booster demonstrated the highest payload capacity among all recorded Falcon 9 launches.

```
%%sql
SELECT Booster_Version,
       PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTABLE
);
```

2015 Launch Records

SQL Query Objective

- List failed drone ship landing attempts during **2015**.

Observation

- The results highlight the technical challenges encountered during the early stages of reusable rocket development.

```
%%sql
SELECT Booster_Version,
       PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTABLE
);
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL Query Objective

- Rank landing outcomes between **2010-06-04** and **2017-03-20**.

Observation

- Landing success improved steadily over the study period, reflecting SpaceX's rapid progress in booster recovery technology.

```
%%sql
SELECT Landing_Outcome,
       COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-
03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
```

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

Global Distribution of Falcon 9 Launch Sites

Objective

- Visualize the geographic locations of all Falcon 9 launch sites using Folium.

Key Observations

- Four primary launch sites were mapped.
- Launch complexes are concentrated along the U.S. coastline.
- Geographic visualization provides spatial context for launch operations.

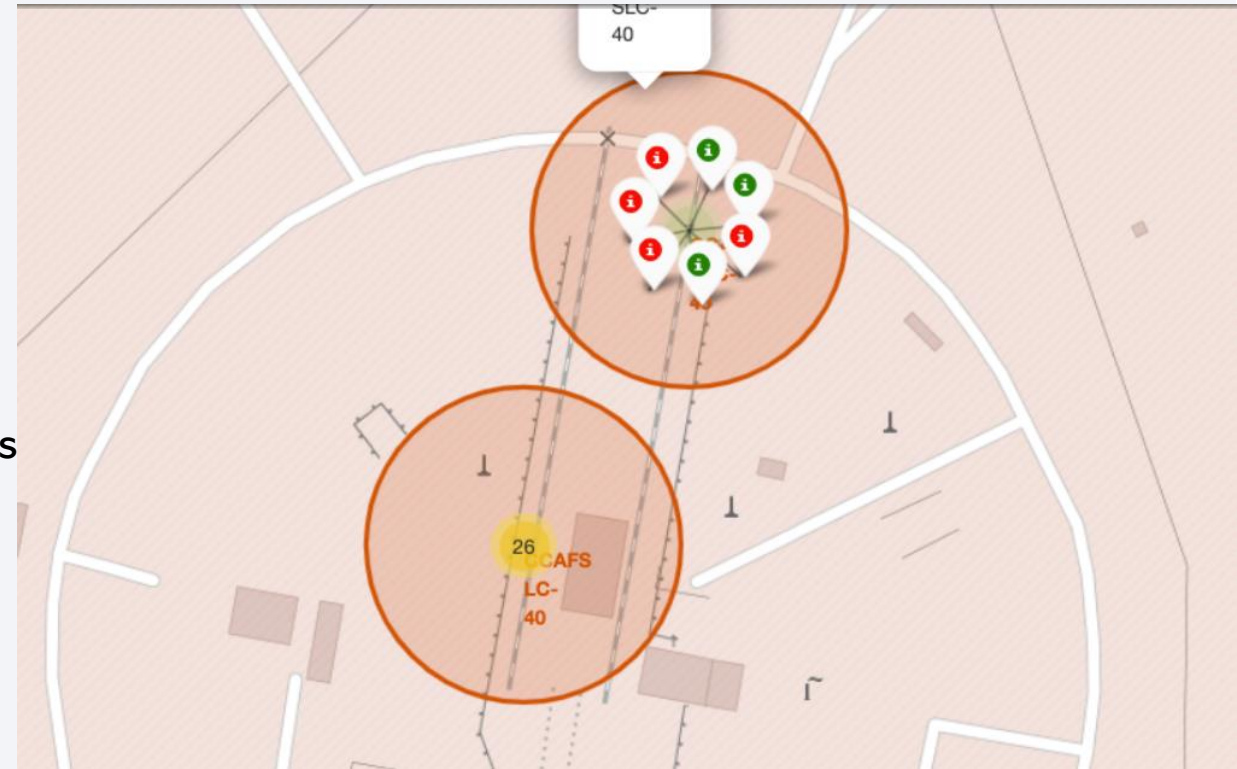
<Folium Map Screenshot 2>

Objective

- Visualize landing outcomes using color-coded map markers.

Key Observations

- Green markers indicate successful landings.
- Red markers represent unsuccessful landing attempts.
- Launch success improved substantially over successive missions



<Folium Map Screenshot 3>

Launch Site Infrastructure Analysis

Objective

- Evaluate the proximity of launch sites to nearby infrastructure.

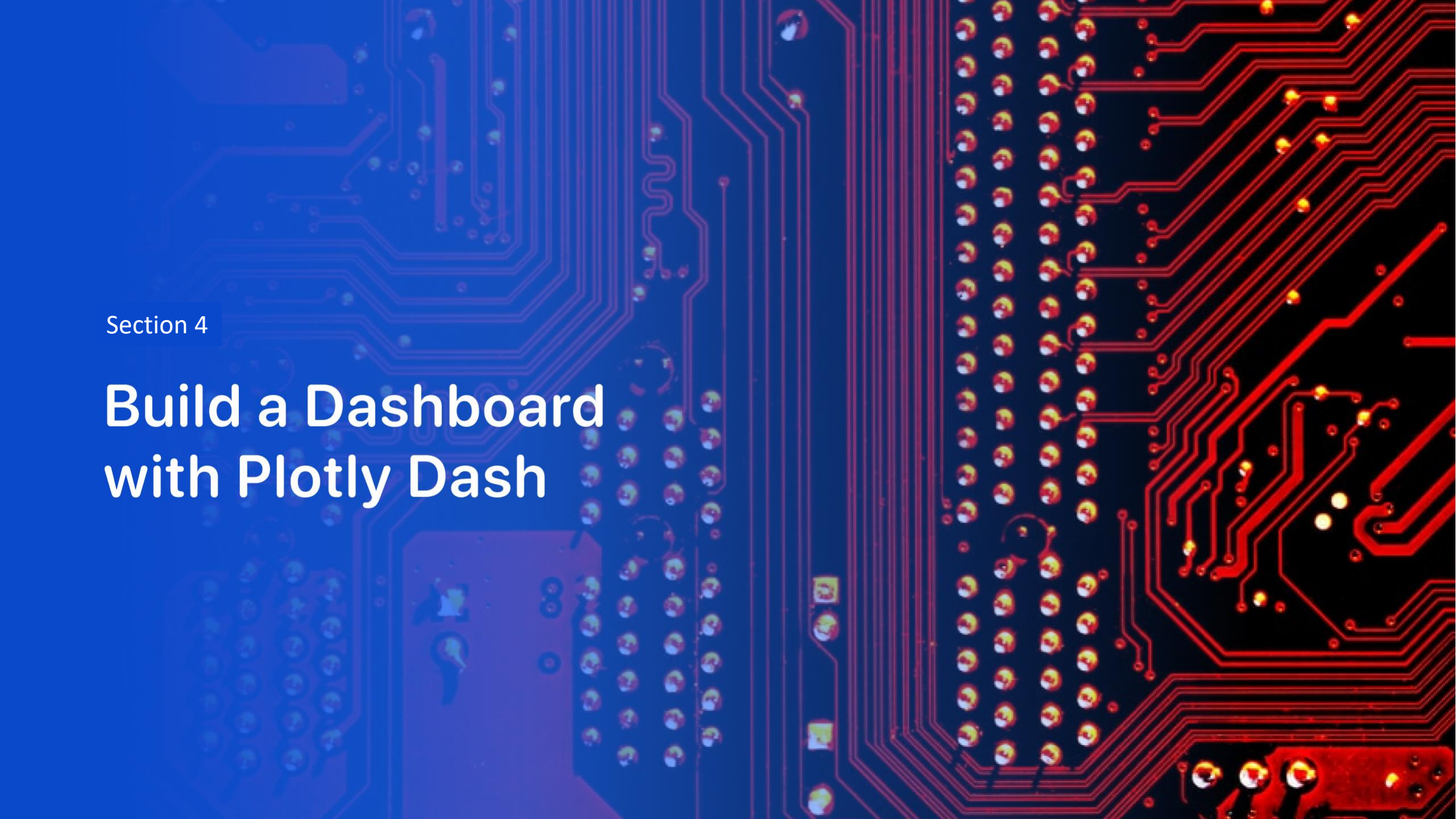
Features Analyzed

- Coastline
- Highway
- Railway
- Airport

Observation

- The launch sites are strategically located near transportation infrastructure while maintaining safe operational distances.





Section 4

Build a Dashboard with Plotly Dash

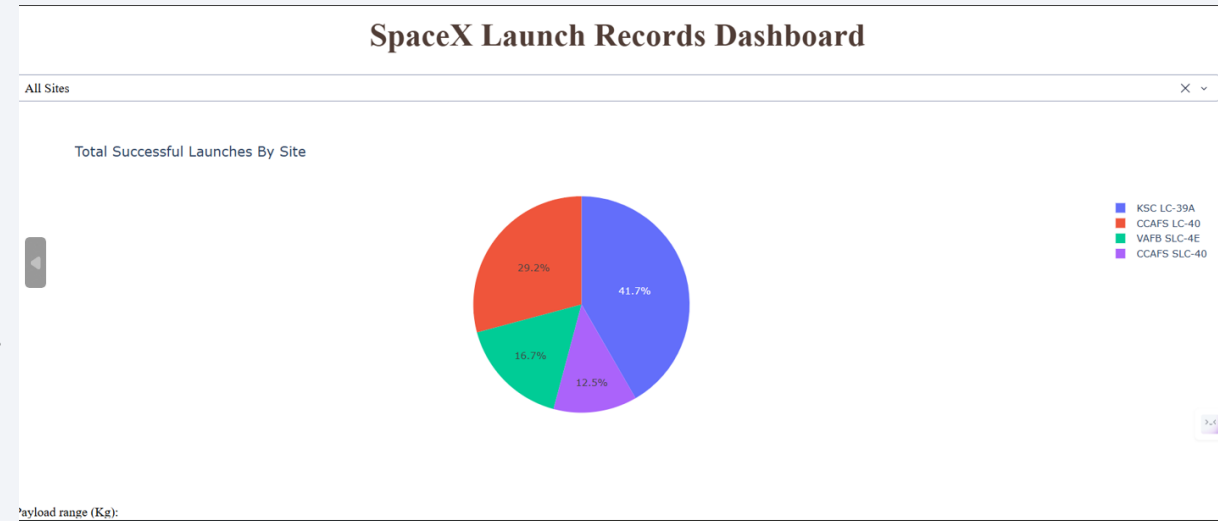
<Dashboard Screenshot 1>

Launch Success by Site

- Objective
- Compare successful launches across all launch sites.

Key Observations

- Launch success varies by launch site.
- Some launch sites recorded consistently higher success rates.
- Interactive filtering allows rapid comparison.



<Dashboard Screenshot 2>

Success Ratio by Launch Site

Objective

Analyze launch success for the selected launch site.

Key Observations

- Certain launch sites achieved higher landing success.
- Pie chart visualization simplifies comparison.
- Success ratio reflects operational maturity.

<Dashboard Screenshot 3>

Payload Mass vs Landing Success

Objective

- Analyze how payload mass influences landing success.

Key Observations

- Payload mass affects landing outcome.
- Payload slider enables interactive analysis.
- Different booster versions perform differently across payload ranges.





Section 5

Predictive Analysis (Classification)

Classification Accuracy

Model Performance Comparison

Model	Test Accuracy
Logistic Regression	83.33%
Support Vector Machine	83.33%
Decision Tree	66.67%
K-Nearest Neighbors	83.33%

Observation

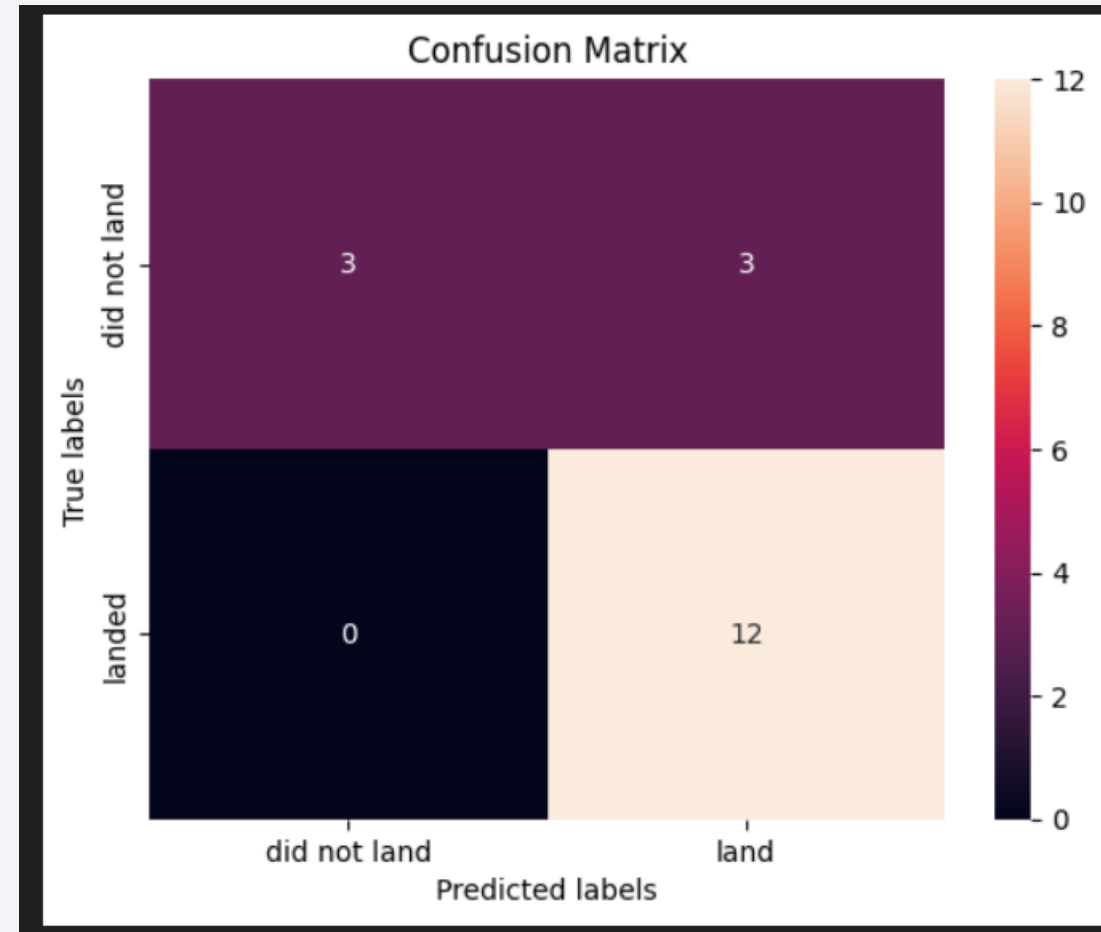
Logistic Regression, SVM, and KNN achieved the highest test accuracy (83.33%), demonstrating that multiple classification algorithms can effectively predict Falcon 9 first-stage landing success.

Confusion Matrix

Model Evaluation

Interpretation

- Most landing outcomes were classified correctly.
- False classifications were limited.
- The confusion matrix demonstrates strong predictive performance.



Conclusions

Project Summary

Key Achievements

- Successfully collected data from multiple sources.
- Performed data cleaning and exploratory analysis.
- Built interactive visualizations using Folium and Plotly Dash.
- Developed and optimized multiple classification models.

Key Findings

- Launch site significantly influences landing success.
- Payload mass and orbit type are important predictive features.
- Machine learning achieved **83.33%** prediction accuracy.

Future Work

- Include weather and telemetry data.
- Explore ensemble learning techniques.
- Evaluate deep learning models for improved prediction.

Appendix

GitHub Repository

<https://github.com/Frqmrg/Frqmrg>

Included Assets

- Python notebooks
- SQL queries
- Machine learning code
- Visualization outputs
- Dashboard implementation
- Supporting datasets

Thank you!

